Supermarket Checkout Simulation Report

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Introduction

Effective queue management at supermarket checkout counters is critical to maintaining customer satisfaction and operational efficiency. This report presents a simulation study of a supermarket with four checkout counters, each operated by a dedicated cashier with distinct service efficiencies. The primary focus is analyzing customer waiting times, cashier utilization, total time customers spend in the supermarket, and service capacity under the assumption that customers choose the shortest queue available upon checkout, thereby providing operational insights into queue management and resource allocation.

Methodology

System Overview

A discrete-event simulation model was developed to represent customer arrivals, shopping times, and checkout processes in a supermarket operating for 13 hours daily (8 AM to 9 PM).

Key features and assumptions include:

Customer Arrivals: Modeled as a Poisson process with an average of 5 customers per minute.

Shopping Times: Generated using a log-normal distribution with parameters $\log(\ \mu) = 0$ and $\log(\ \sigma) = 0.5$.

Checkout Process:

Four dedicated cashiers with service times exponentially distributed with rate parameters corresponding to mean service times.

- Counter 1: μ=1/2.5 (2.5 mins per customer)
- Counter 2: μ=1/3 (3 mins per customer)
- Counter 3: μ=1/3.5 (3.5 mins per customer)
- Counter 4: μ=1/4 (4 mins per customer)

Customers select the shortest queue among the four counters on arriving at checkout and ties are broken randomly.

Service Completion: Customers arriving before 9:00 p.m. can complete service regardless of finish time

Tools Used

Language: R

Random Variable Generation:

- rexp() for exponential service times
- rlnorm() for log-normal shopping times

Data Structures: Data frames and Lists

Key Variables

- **Event List:** Next customer arrival time, next service completion at each counter.
- **System States:** Number of customers currently waiting in each queue, Cashier Status; Busy or idle, Total number of customers currently in the supermarket
- **Counter Variables:** Number of arrivals by time t, Number of departures after service completion by time t
- **Customer States:** Arrival time, shopping end time, assigned counter, Waiting time, checkout start time, checkout end time, Total time
- **Queue Lengths:** Number of customers waiting at each counter.

Events

- **Customer Arrival:** A new customer enters, shops, and joins the shortest checkout queue.
- **Shopping Completion:** Customer finishes shopping and joins the checkout queue.
- **Service Start:** Customer begins checkout if the cashier is idle.
- **Service Completion:** Customer leaves the checkout; the next in queue is served.

Simulation Approach

The simulation is replicated 100 times (simulated days) with different random seeds to capture variability and produce robust statistical summaries, including 95% confidence intervals for key metrics.

Results and Discussion

Simulation outputs averaged across 100 replications are as follows:

Counter	Mean_Avg_Waiting_Time	Mean_Utilization
1	902.653	0.701
2	1087.485	0.802
3	1268.845	0.900
4	1463.418	0.999

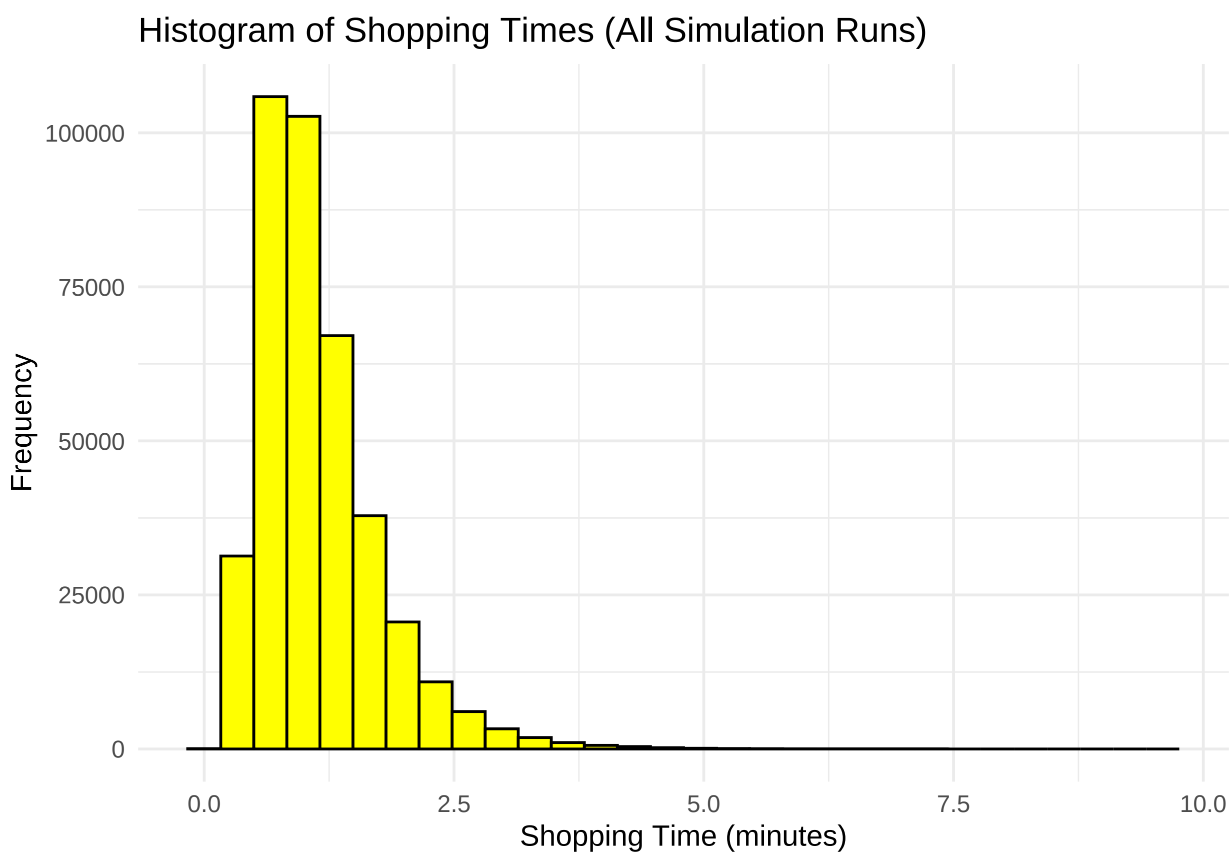
```
##
## Average total time in supermarket: 1175.13 minutes (95% CI: 1093.68 - 1246.80)

## Average number of customers served per day: 3900.88 (95% CI: 3772.80 - 4017.90)
```

Graph Interpretations

Histogram of Shopping Times

The distribution of shopping times across all simulation runs reveals that most customers spend between 0.5 to 3 minutes shopping, with a right-skewed distribution.

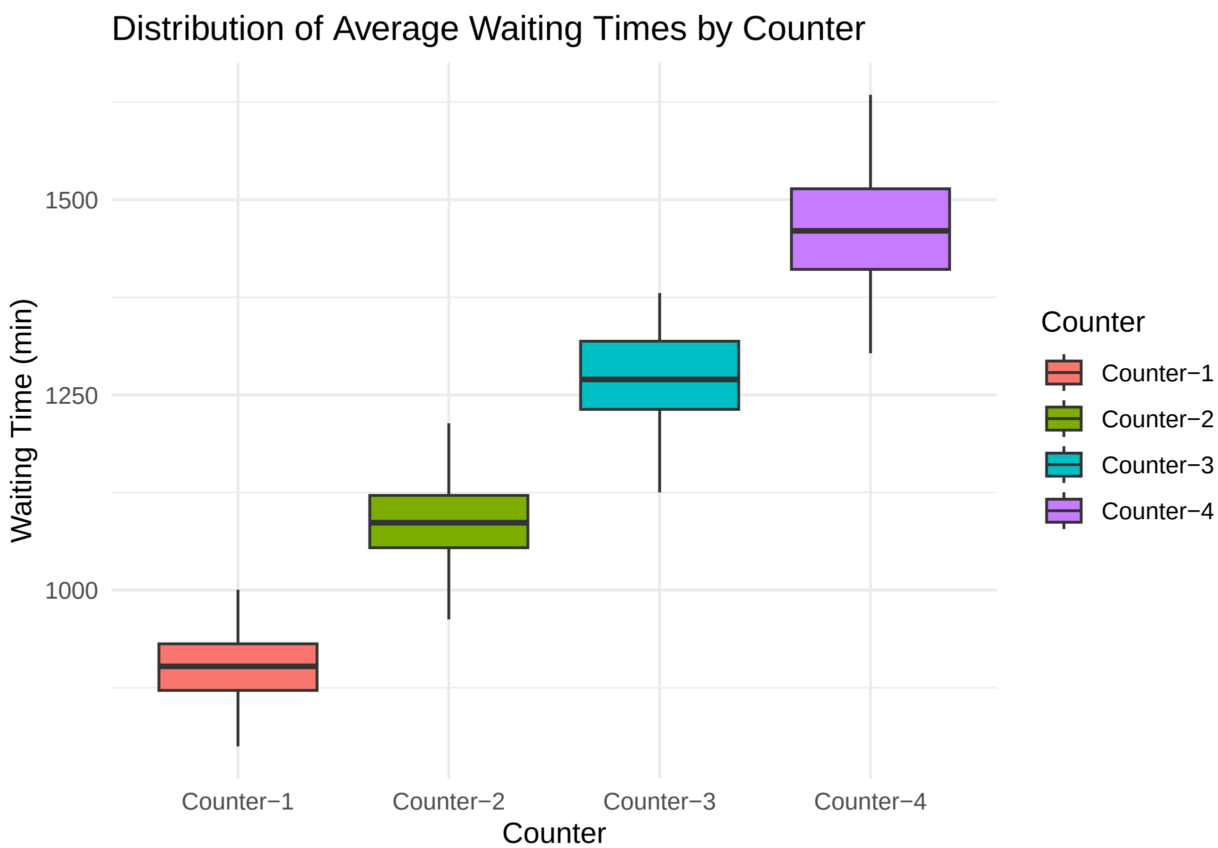
The very short shopping times mean customers quickly move from shopping to checkout, contributing to rapid queue formation at checkout counters.



Distribution of Average Waiting Times by Counter

Box plots comparing waiting time distributions across the four counters, showing increasing median waiting times and variability from Counter 1 to Counter 4.

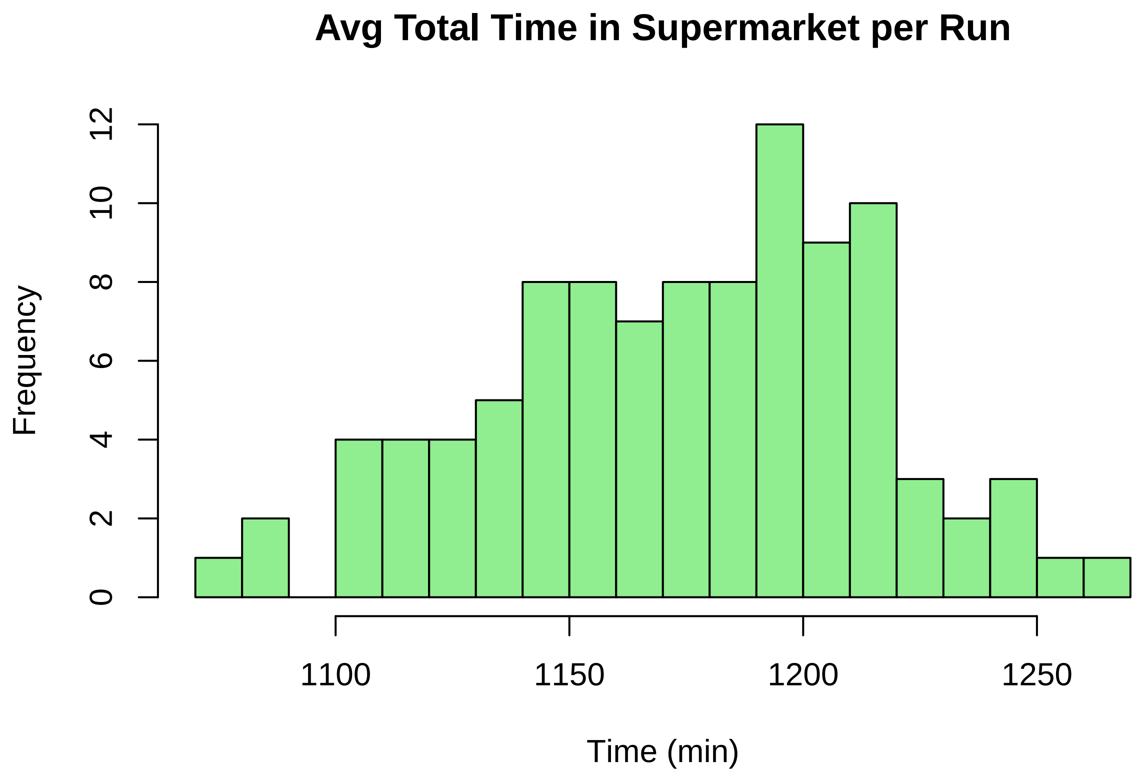
Even the "fastest" counter (Counter 1) has extremely long waits, while Counter 4 shows the highest and most variable waiting times, indicating severe congestion at slower service points.



Average Total Time in Supermarket per Run

This histogram displays the variation in average total customer time across the 100 simulation runs, clustering around 1150-1200 minutes.

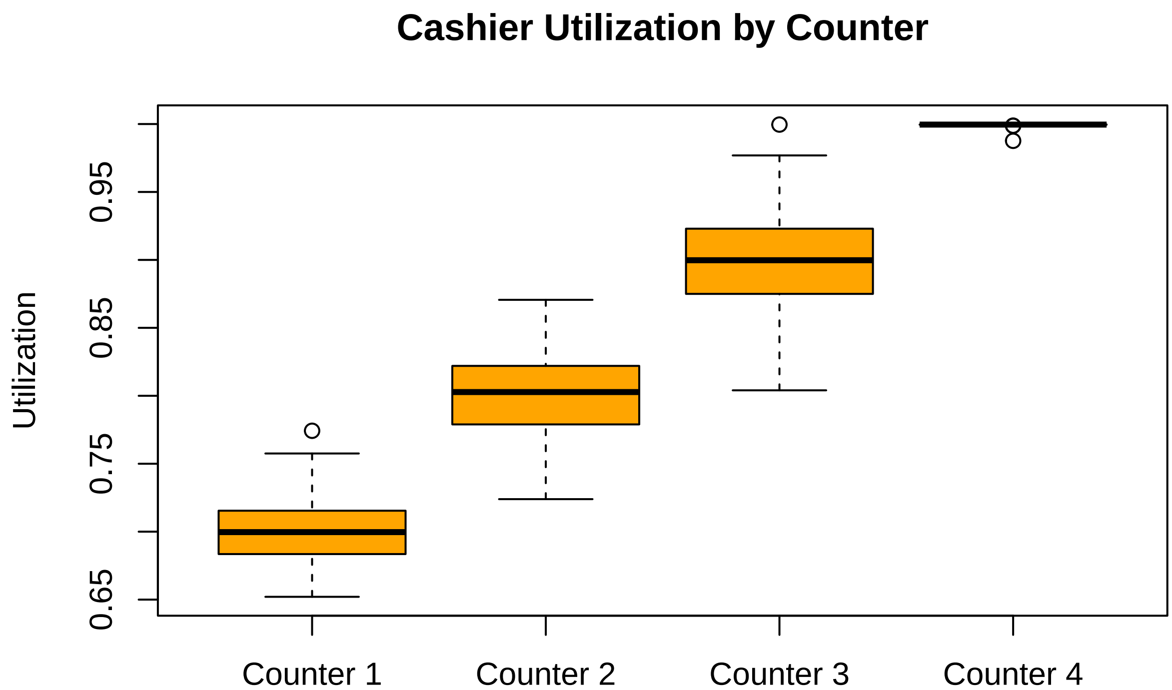
The tight distribution across runs confirms that the system is reliably overloaded, with customers consistently spending unrealistic amounts of time in the store.



Cashier Utilization by Counter

Box plots displaying utilization rates for each counter, showing increasing utilization from Counter 1 (around 70%) to Counter 4 (nearly 100%).

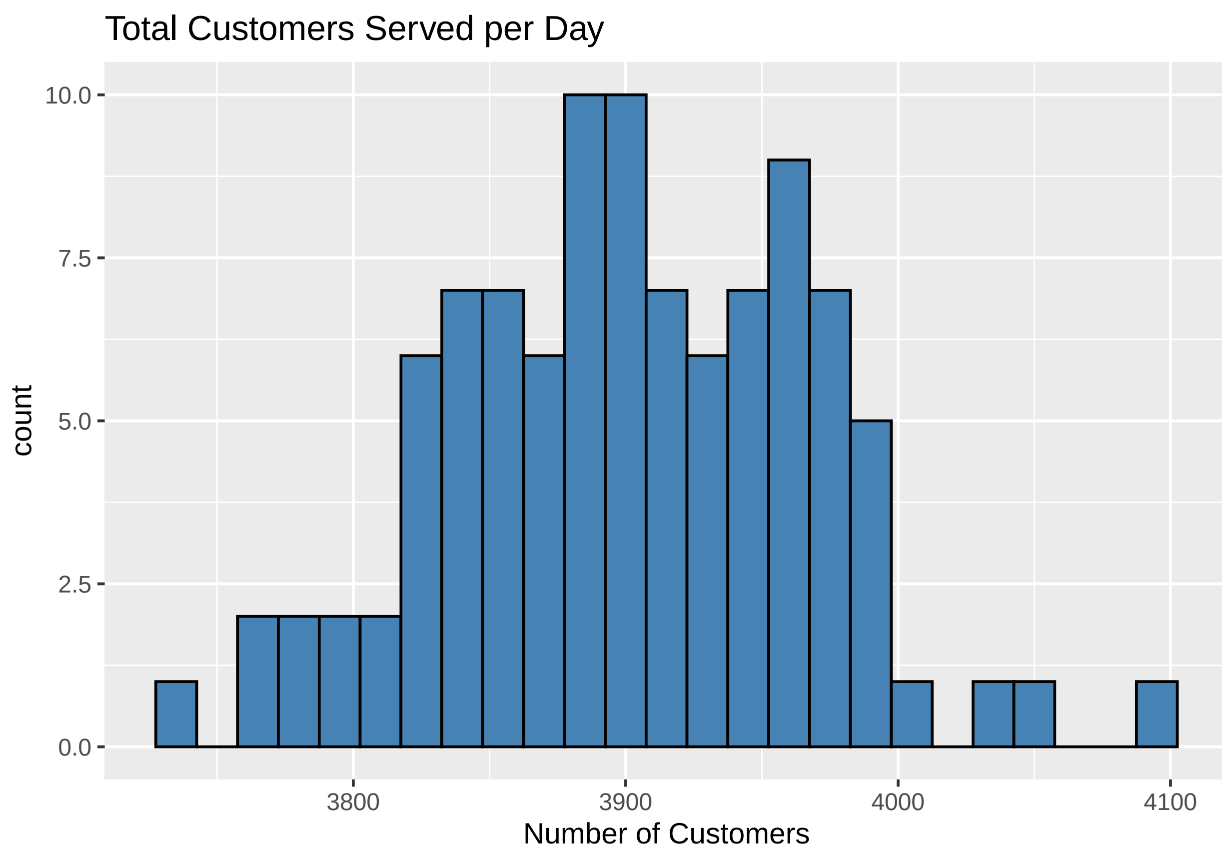
Higher utilization at slower counters indicates these resources are working at maximum capacity, creating bottlenecks that contribute significantly to overall system congestion.



Total Customers Served per Day

The distribution of daily customer volumes across simulation runs, centered around 3900 customers with relatively low variability.

Despite the high arrival rate successfully bringing in many customers, the system cannot process them efficiently, leading to the observed long waiting times.



Key Findings

- **Excessive Waiting Times:** The simulation reveals extremely long waiting times ranging from approximately 15 hours at the fastest counter to over 24 hours at the slowest counter. These figures are unrealistic compared to typical supermarket experiences where waiting times are usually measured in minutes.
- **High Cashier Utilization:** Utilization rates increase progressively from 70% at Counter 1 to nearly 100% at Counter 4. The near-maximum utilization indicates cashiers are constantly busy, creating bottlenecks and extended queues.
- **System Overload:** The combination of high arrival rates (5 customers per minute) and short shopping times (mostly 1-3 minutes) creates rapid queue buildup that overwhelms the available service capacity.

Conclusion

The simulation shows that the current supermarket setup is unstable and not practical for real-world use. The main issues are very short shopping times (1–3 minutes) combined with a high arrival rate of 5 customers per minute, creating a bottleneck at checkout. The available cashiers cannot handle the fast influx of customers, leading to extremely long waiting times of 15 to 24 hours and nearly full cashier utilization. While choosing the shortest queue helps distribute customers, it doesn't solve the underlying capacity problem.

To fix this and improve customer experience, the supermarket needs to increase service capacity by adding more checkout counters, express lanes, or self-checkouts. It should also optimize customer flow by encouraging longer shopping times and managing peak arrivals. Improving service efficiency through cashier training and technology can help speed things up. This simulation highlights how important it is to balance customer demand with available service resources and can guide better decisions on staffing and store layout for a smoother shopping experience.